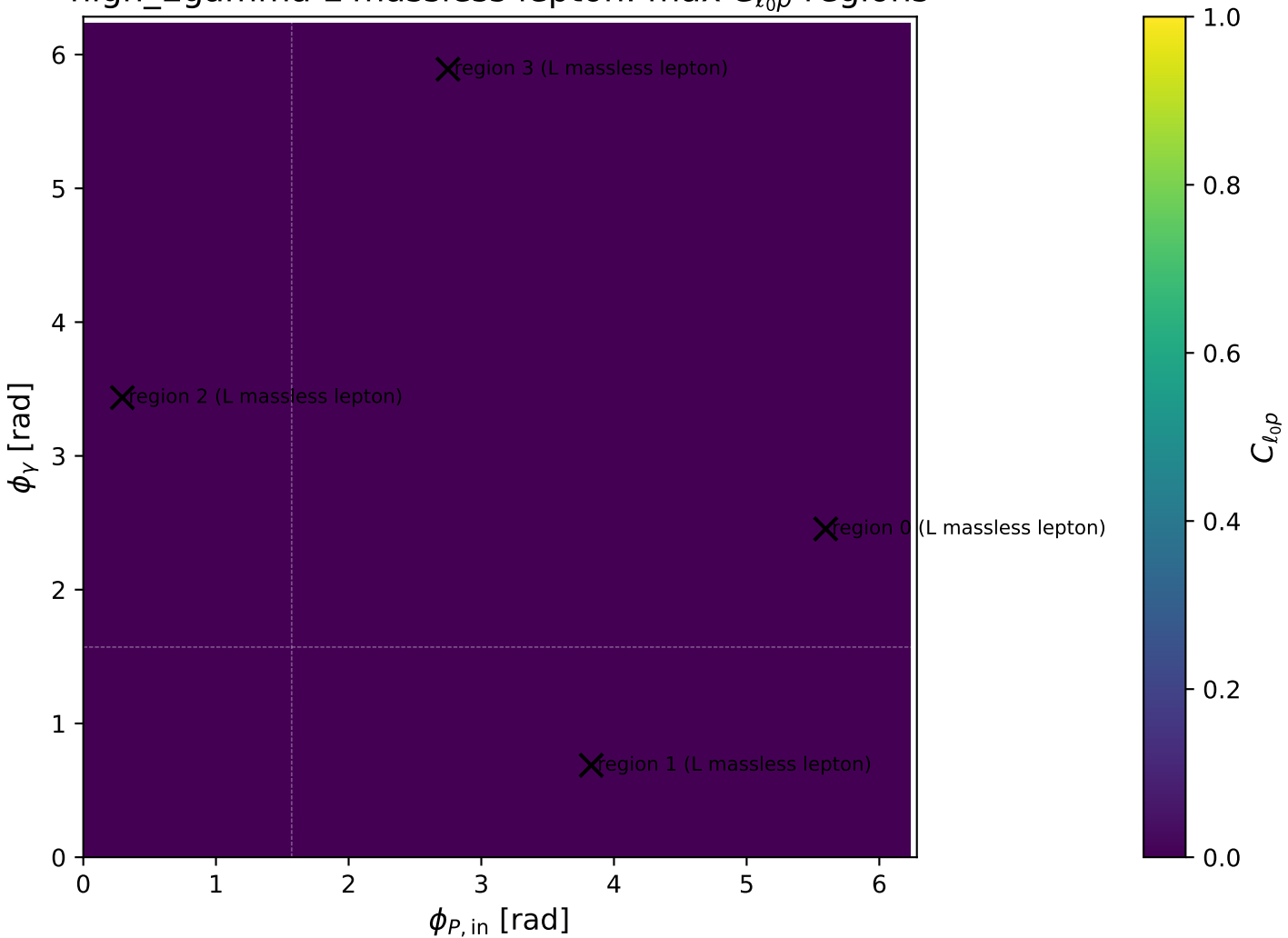
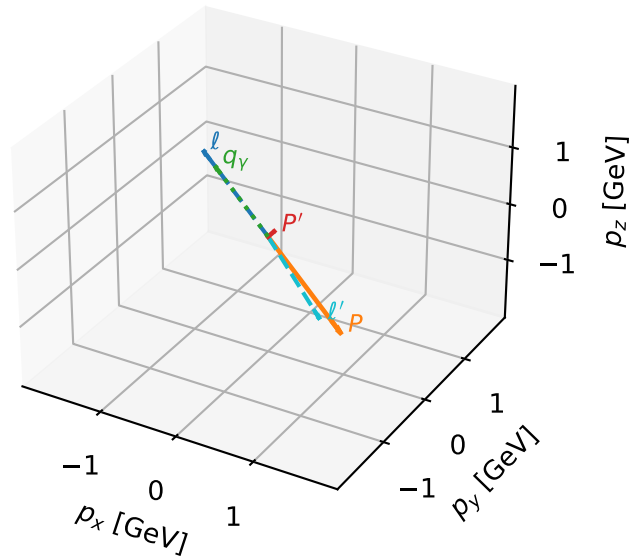


high\_Egamma L massless lepton: max  $C_{\ell_0 p}$  regions



# L massless lepton characteristic max $C_{\ell_0 p}$ configuration

3D momenta

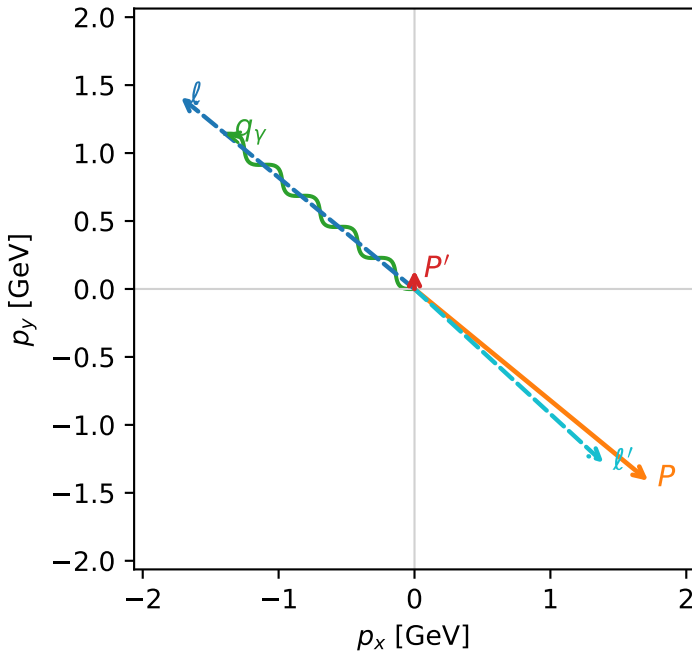


c\_ep\_high\_egamma\_l\_electron\_region\_0  $C_{\{\ell_0 p\}}=3.07555e-12$   
 region: high\_Egamma\_L\_electron\_region\_0 final-pair delta\_xy=2.31441 rad  
 outgoing-state purity: 0.999986  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

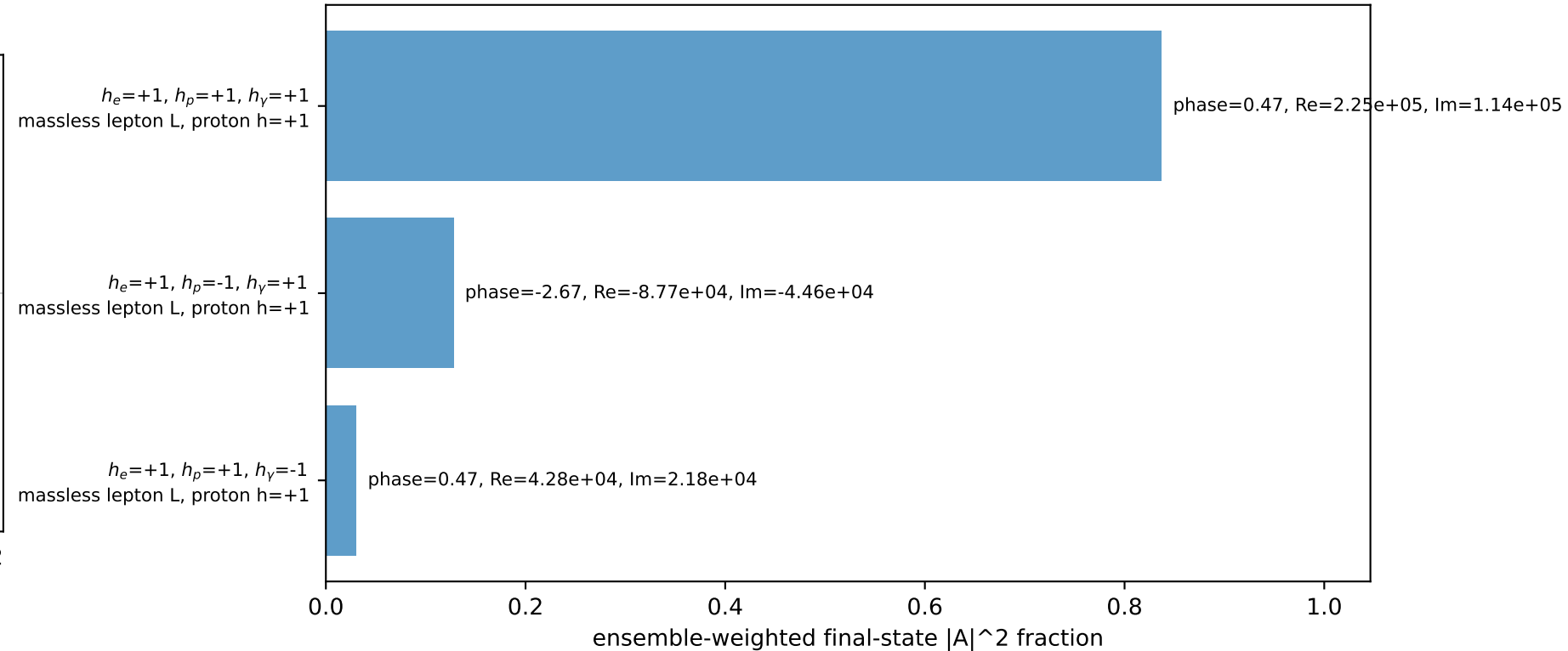
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.137832$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=2.45437$ ,  $\phi_P=5.59596$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=2.45437$   
 $Q^2=16.8072$ ,  $x_B=0.84435$ ,  $t=-3.2068$ ,  $W^2=3.97812$ ,  $y=0.964599$   
 $\theta(\ell', \gamma)=176.769$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -1.7195$   $p_y= 1.4112$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 1.7195$   $p_y= -1.4112$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8904$   $p_x= 1.3914$   $p_y= -1.2797$   $p_z= 0.0000$   
 $P'$   $E= 0.9481$   $p_x= 0.0000$   $p_y= 0.1378$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.3914$   $p_y= 1.1419$   $p_z= 0.0000$

Transverse plane

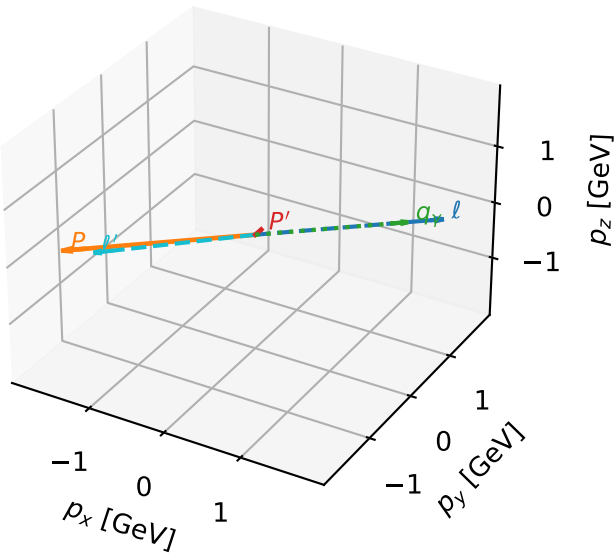


Leading initial-ensemble/final-state components (N=3, retained=99.5%)



# L massless lepton characteristic max $C_{\ell_0 p}$ configuration

3D momenta

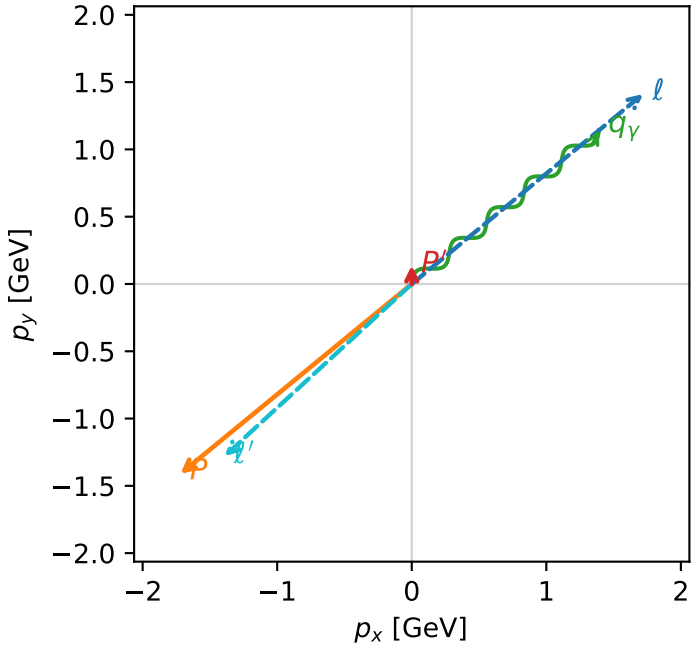


c\_ep\_high\_egamma\_l\_electron\_region\_1 C\_{\ell\_0 p}=2.77638e-12  
region: high\_Egamma\_L\_electron\_region\_1 final-pair delta\_xy=2.31441 rad  
outgoing-state purity: 0.999986  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

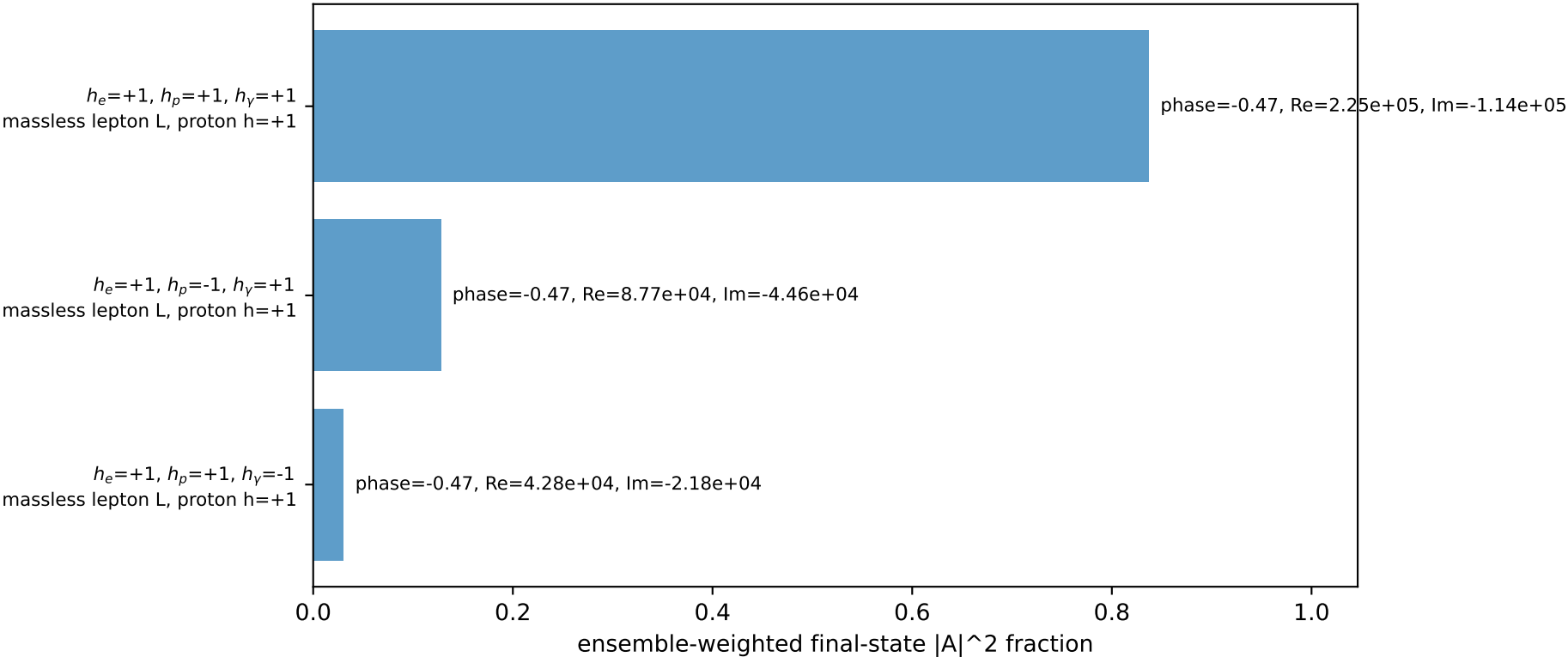
s=21.5158,  $\sqrt{s}$ =4.63852,  $|\vec{P}|$ =2.22442,  $|\vec{P}'|$ =0.137832  
 $\theta_{in}$ =1.57079,  $\phi_{\ell}$ =0.687223,  $\phi_P$ =3.82882  
 $E_V$ =1.8,  $\phi_V$ =0.687223  
 $Q^2$ =16.8072,  $x_B$ =0.84435,  $t$ =-3.2068,  $W^2$ =3.97812,  $y$ =0.964599  
 $\theta(\ell', \gamma)$ =176.769 deg

four-momenta [E, p\_x, p\_y, p\_z] GeV:  
 $\ell$  E= 2.2244 p\_x= 1.7195 p\_y= 1.4112 p\_z= -0.0000  
P E= 2.4141 p\_x= -1.7195 p\_y= -1.4112 p\_z= 0.0000  
 $\ell'$  E= 1.8904 p\_x= -1.3914 p\_y= -1.2797 p\_z= 0.0000  
P' E= 0.9481 p\_x= 0.0000 p\_y= 0.1378 p\_z= 0.0000  
 $q_\gamma$  E= 1.8000 p\_x= 1.3914 p\_y= 1.1419 p\_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=99.5%)

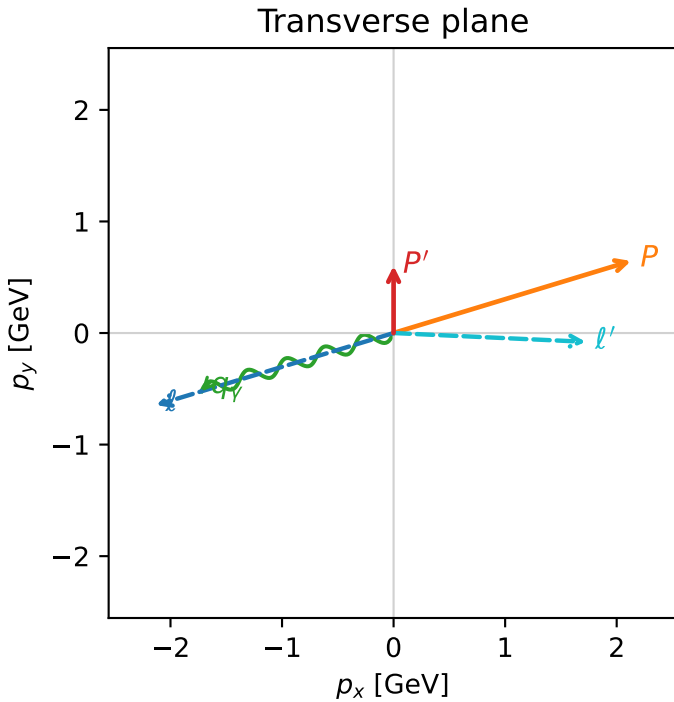
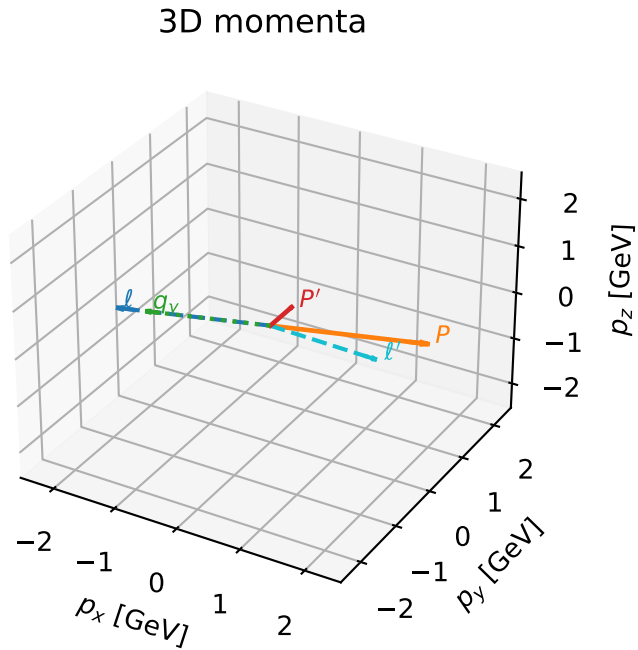


# L massless lepton characteristic max $C_{\ell_0 p}$ configuration

c\_ep\_high\_egamma\_l\_electron\_region\_2  $C_{\{\ell_0 p\}}=1.73244\text{e-}12$   
region: high\_Egamma\_L\_electron\_region\_2 final-pair delta\_xy=1.61654 rad  
outgoing-state purity: 0.984594  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.601368$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=3.43612$ ,  $\phi_P=0.294524$   
 $E_{\gamma}=1.8$ ,  $\phi_{\gamma}=3.43612$   
 $Q^2=14.9024$ ,  $x_B=0.762581$ ,  $t=-2.84337$ ,  $W^2=5.51948$ ,  $y=0.946987$   
 $\theta(\ell', \gamma)=160.504$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -2.1286$   $p_y= -0.6457$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 2.1286$   $p_y= 0.6457$   $p_z= 0.0000$   
 $\ell'$   $E= 1.7243$   $p_x= 1.7225$   $p_y= -0.0789$   $p_z= 0.0000$   
 $P'$   $E= 1.1142$   $p_x= 0.0000$   $p_y= 0.6014$   $p_z= 0.0000$   
 $q_{\gamma}$   $E= 1.8000$   $p_x= -1.7225$   $p_y= -0.5225$   $p_z= 0.0000$

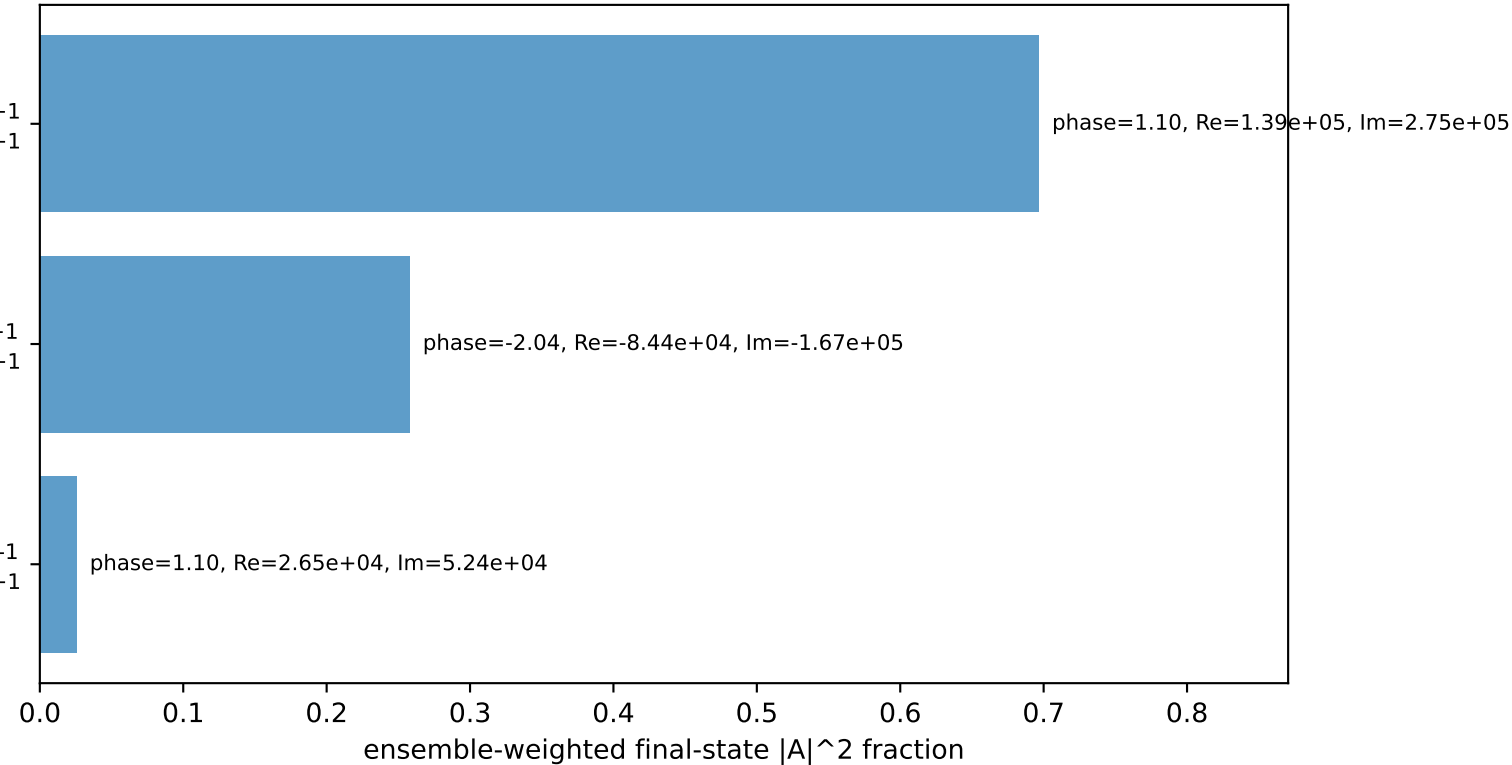


$h_e=+1, h_p=+1, h_{\gamma}=+1$   
massless lepton L, proton  $h=+1$

$h_e=+1, h_p=-1, h_{\gamma}=+1$   
massless lepton L, proton  $h=+1$

$h_e=+1, h_p=+1, h_{\gamma}=-1$   
massless lepton L, proton  $h=+1$

Leading initial-ensemble/final-state components (N=3, retained=97.9%)



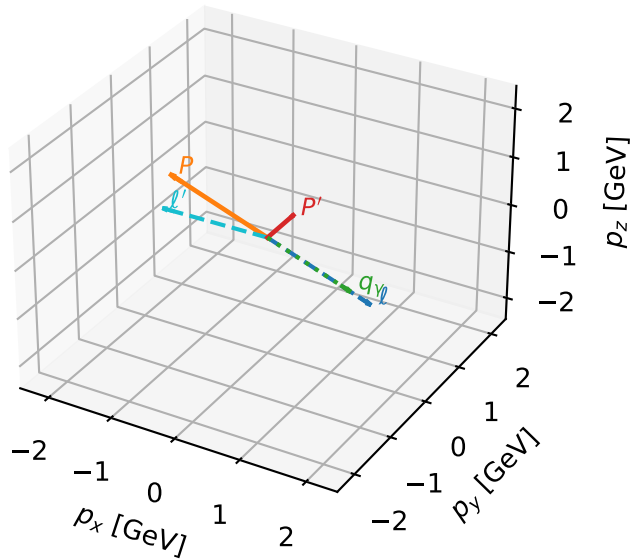
# L massless lepton characteristic max $C_{\ell_0 p}$ configuration

c\_ep\_high\_egamma\_l\_electron\_region\_3  $C_{\ell_0 p}=9.62919\text{e-}13$   
 region: high\_Egamma\_L\_electron\_region\_3 final-pair delta\_xy=1.58254 rad  
 outgoing-state purity: 0.971774  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

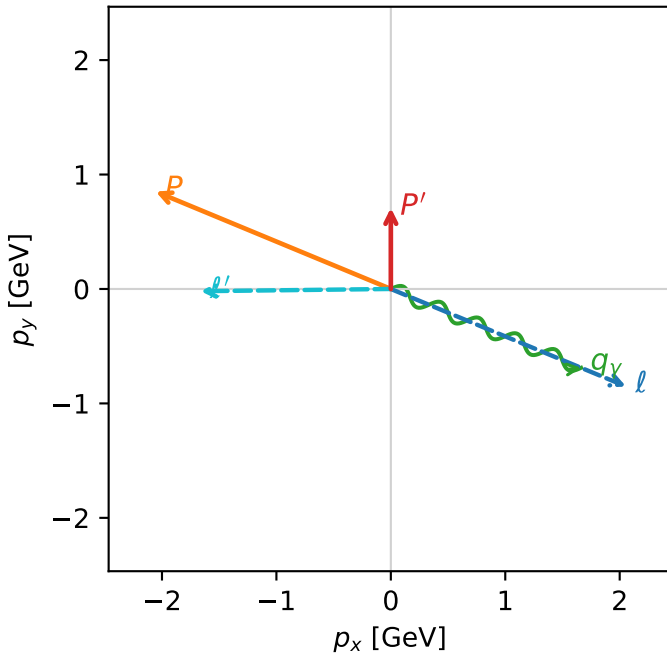
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.708355$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=5.89049$ ,  $\phi_P=2.74889$   
 $E_Y=1.8$ ,  $\phi_Y=5.89049$   
 $Q^2=14.2008$ ,  $x_B=0.731691$ ,  $t=-2.7095$ ,  $W^2=6.08723$ ,  $y=0.9405$   
 $\theta(\ell', \gamma)=156.827$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 2.0551$   $p_y= -0.8512$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.0551$   $p_y= 0.8512$   $p_z= 0.0000$   
 $\ell'$   $E= 1.6631$   $p_x= -1.6630$   $p_y= -0.0195$   $p_z= 0.0000$   
 $P'$   $E= 1.1754$   $p_x= 0.0000$   $p_y= 0.7084$   $p_z= 0.0000$   
 $q_Y$   $E= 1.8000$   $p_x= 1.6630$   $p_y= -0.6888$   $p_z= 0.0000$

3D momenta



Transverse plane



Leading initial-ensemble/final-state components (N=3, retained=97.2%)

